

# Antihypertensive agents

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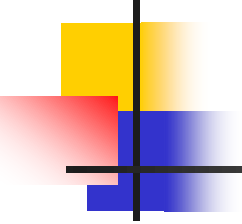
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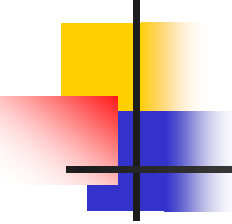
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# วัตถุประสงค์การเรียนรู้

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- อธิบายแนวทางในการรักษาความดันโลหิตสูง
- บอกชื่อยาหลักและกลุ่มยาที่ใช้รักษาความดันโลหิตสูง
- อธิบายกลไกการออกฤทธิ์และเภสัชวิทยาของยาที่ใช้รักษาความดันโลหิตสูง
- เลือกใช้ยารักษาความดันโลหิตสูงได้อย่างเหมาะสม

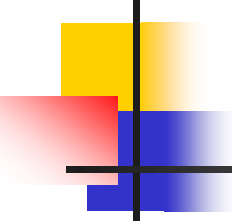


# Classification of hypertension

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## PRIMARY HYPERTENSION

- ◉ No medical cause
- ◉ Risk factors :
  - Sedentary lifestyle
  - Obesity
  - Salt (sodium) sensitivity
  - Alcohol, smoking
  - Family history



# Classification of hypertension

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## SECONDARY HYPERTENSION

- Adrenal cortical abnormalities :
  - Cushing's syndrome ( adrenal glands overproduce the hormone cortisol)
  - Primary aldosteronism ( overproduction of aldosterone by adrenal glands)
- Diseases of the kidney (polycystic kidney disease – genetic disorder of the kidneys)
- Diseases of the renal arteries supplying the kidney – renovascular hypertension
- Neuroendocrine tumors (pheochromocytoma)



## การจำแนกความรุนแรงของระดับความดันโลหิตตามระดับความดันโลหิตที่วัดได้จาก สถานพยาบาลในบุคคลที่มีอายุตั้งแต่ 18 ปีขึ้นไป

| การวินิจฉัย                              | ค่าความดันโลหิตซิสโตลิก<br>(มม. ปรอท) |          | ค่าความดันโลหิต<br>ไดแอสโตลิก (มม.<br>ปรอท) |
|------------------------------------------|---------------------------------------|----------|---------------------------------------------|
| Optimal                                  | น้อยกว่า 120                          | และ      | น้อยกว่า 80                                 |
| Normal                                   | 120 ถึง 129                           | และ/หรือ | น้อยกว่า 80                                 |
| BP at risk                               | 130 ถึง 139                           | และ/หรือ | 80 ถึง 89                                   |
| ความดันโลหิตสูงระดับ 1                   | 140 ถึง 159                           | และ/หรือ | 90 ถึง 99                                   |
| ความดันโลหิตสูงระดับ 2                   | 160 ถึง 179                           | และ/หรือ | 100 ถึง 109                                 |
| ความดันโลหิตสูงระดับ 3                   | ตั้งแต่ 180 ขึ้นไป                    | และ/หรือ | ตั้งแต่ 110 ขึ้นไป                          |
| Isolated systolic<br>hypertension (ISH)  | ตั้งแต่ 140 ขึ้นไป                    | และ      | น้อยกว่า 90                                 |
| Isolated diastolic<br>hypertension (IDH) | น้อยกว่า 140                          | และ      | ตั้งแต่ 90 ขึ้นไป                           |

# The cardiovascular continuum in hypertension and other risk factors

## Eye

- Microvascular remodelling
- Hypertensive retinopathy

## Brain

- White matter lesions
- Silent microinfarcts
- Microbleeds
- Brain atrophy
- Cognitive impairment
- Vascular dementia
- Ischaemic stroke
- Cerebral haemorrhage

## Heart

- LVH
- LA and LV dilatation
- AF
- Obstructive and non-obstructive CAD
- Myocardial Infarction
- Diastolic and/or systolic heart failure

## Kidney

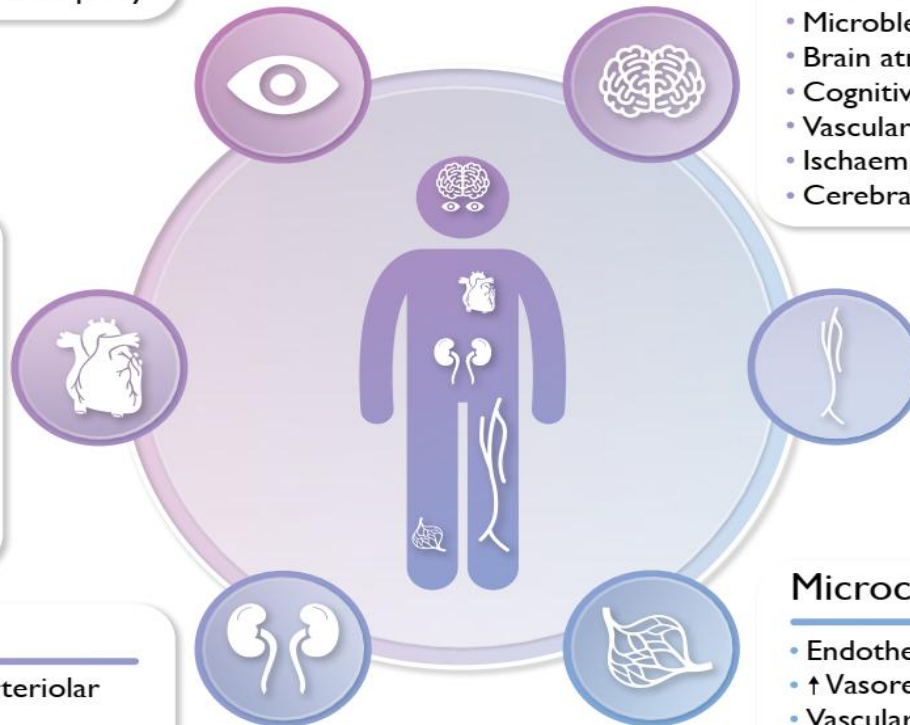
- Glomerular arteriolar hypertension
- Glomerulosclerosis
- Albuminuria/Proteinuria
- ↓ GFR

## Large and medium arteries

- Atherosclerosis
- Vascular calcification
- Arterial stiffness

## Microcirculation

- Endothelial dysfunction
- ↑ Vasoreactivity
- Vascular remodelling
- Fibrosis and inflammation
- ↑ Peripheral vascular resistance



See footnote for information on sex-differences

# Hypertension Guidelines

## **2024 ESH/ESC**

<https://www.portailvasculaire.fr/sites/default/files/docs/ehae178.pdf>

## ■ **2024 Thai Guideline**

<https://www.thaihypertension.org/files/Guideline2024.pdf>

## ■ **2025 ACC/AHA**

<https://www.jacc.org/doi/epdf/10.1016/j.jacc.2025.05.007>

# Treatment goal

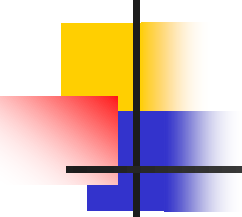
“Reduce morbidity and mortality”

## 2025 ACC/AHA

**BP Goal <130/80**

## 2024 ESH/ 2024 Thai Guideline

- **BP Goal <140/90 in all pts**
  - If the treatment is well tolerated, BP Goal <130/80 in most pts
  - Treated SBP should not be targeted to <120 mmHg
    - Pts with age <65 SBP target 120-129
    - Pts with age ≥65 SBP target 130-139
  - Suitable DBP: 70-79 mmHg



# Management of Hypertension

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1. **Lifestyle modification:** decreased BP, enhanced drug efficacy and decreased cardiovascular risks
  - No smoking
  - Weight control
  - Exercise
  - Reduce alcohol intake:  $\leq 30$  cc per day
  - Decrease stress
  - Sodium control:  $\leq 1 - 2$  g per day
  - Limit caffeine consumption
  - Maintain adequate intake of  $\text{Ca}^{2+}$ ,  $\text{K}^{+}$ ,  $\text{Mg}^{2+}$

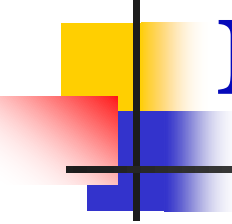
# Best Proven Nonpharmacological Interventions for Prevention and Treatment of Hypertension\*

|                                             | Non-pharmacological Intervention | Dose                                                                                                                                                                                   | Approximate Impact on SBP |              |
|---------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------|
|                                             |                                  |                                                                                                                                                                                        | Hypertension              | Normotension |
| <b>Weight loss</b>                          | Weight/body fat                  | Best goal is ideal body weight, but aim for at least a 1-kg reduction in body weight for most adults who are overweight. Expect about 1 mm Hg for every 1-kg reduction in body weight. | -5 mm Hg                  | -2/3 mm Hg   |
| <b>Healthy diet</b>                         | DASH dietary pattern             | Consume a diet rich in fruits, vegetables, whole grains, and low-fat dairy products, with reduced content of saturated and total fat.                                                  | -11 mm Hg                 | -3 mm Hg     |
| <b>Reduced intake of dietary sodium</b>     | Dietary sodium                   | Optimal goal is <1500 mg/d, but aim for at least a 1000-mg/d reduction in most adults.                                                                                                 | -5/6 mm Hg                | -2/3 mm Hg   |
| <b>Enhanced intake of dietary potassium</b> | Dietary potassium                | Aim for 3500–5000 mg/d, preferably by consumption of a diet rich in potassium.                                                                                                         | -4/5 mm Hg                | -2 mm Hg     |

# Best Proven Nonpharmacological Interventions for Prevention and Treatment of Hypertension\* (cont.)

|                                     | Nonpharmacological Intervention | Dose                                                                                                                                                                             | Approximate Impact on SBP |              |
|-------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------|
|                                     |                                 |                                                                                                                                                                                  | Hypertension              | Normotension |
| <b>Physical activity</b>            | Aerobic                         | <ul style="list-style-type: none"> <li>● 90–150 min/wk</li> <li>● 65%–75% heart rate reserve</li> </ul>                                                                          | -5/8 mm Hg                | -2/4 mm Hg   |
|                                     | Dynamic resistance              | <ul style="list-style-type: none"> <li>● 90–150 min/wk</li> <li>● 50%–80% 1 rep maximum</li> <li>● 6 exercises, 3 sets/exercise, 10 repetitions/set</li> </ul>                   | -4 mm Hg                  | -2 mm Hg     |
|                                     | Isometric resistance            | <ul style="list-style-type: none"> <li>● 4 × 2 min (hand grip), 1 min rest between exercises, 30%–40% maximum voluntary contraction, 3 sessions/wk</li> <li>● 8–10 wk</li> </ul> | -5 mm Hg                  | -4 mm Hg     |
| <b>Moderation in alcohol intake</b> | Alcohol consumption             | In individuals who drink alcohol, reduce alcohol <sup>†</sup> to: <ul style="list-style-type: none"> <li>● Men: ≤2 drinks daily</li> <li>● Women: ≤1 drink daily</li> </ul>      | -4 mm Hg                  | -3 mm        |

one “standard” drink contains roughly 14 g of pure alcohol, which is typically found in 12 oz of regular beer (usually about 5% alcohol), 5 oz of wine (usually about 12% alcohol), and 1.5 oz of distilled spirits (usually about 40% alcohol).

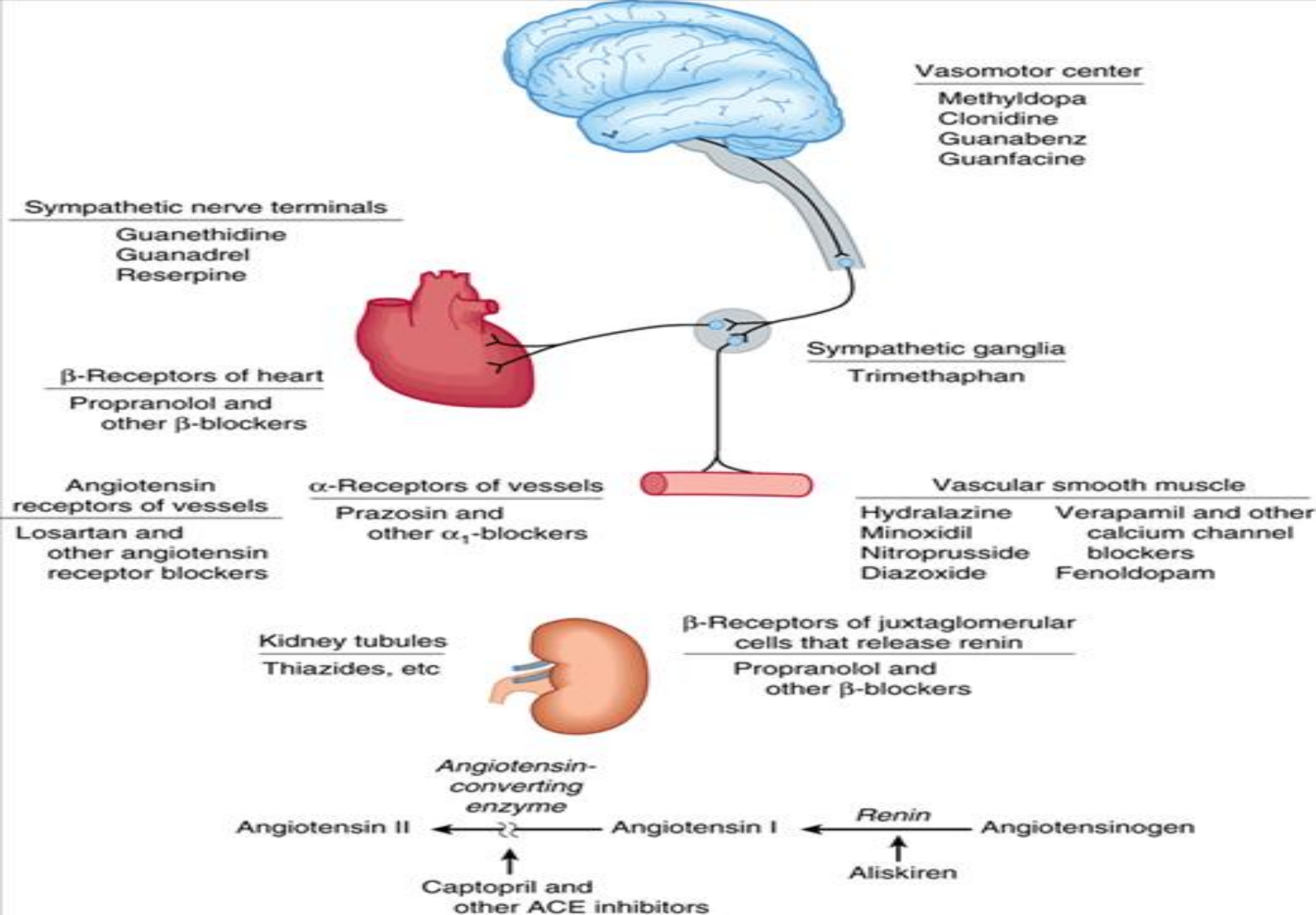


# Management of Hypertension

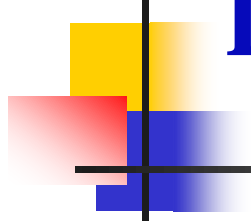
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## 2. Pharmacological treatment

- Diuretics
- Sympatholytic agents
- Direct acting vasodilators
- Calcium channel blockers (CCBs)
- Renin angiotensin aldosterone inhibitors

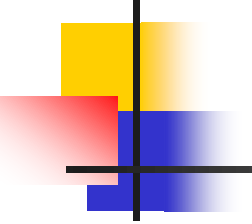


# 1. Diuretics (Thiazide/thiazide-like diuretics)



## Mechanism of action

- Inhibited  $\text{Na}^+\text{-Cl}^-$  symports  $\longrightarrow$   $\downarrow$  body  $\text{Na}^+$  stores  
 $\longrightarrow$   $\downarrow$  blood volume  $\longrightarrow$   $\downarrow$  venous return  $\longrightarrow$   $\downarrow$  stroke  
volume  $\longrightarrow$   $\downarrow$  CO  $\longrightarrow$   $\downarrow$  BP
- Direct vasodilation  $\longrightarrow$   $\downarrow$  peripheral vascular resistance



# Diuretics

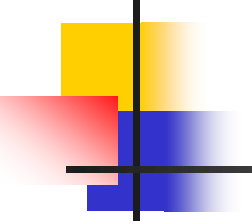
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- Type of Diuretics

- Thiazide diuretics: hydrochlorothiazide (HCTZ), Chlorthalidone, Indapamide
- Loop diuretics: furosemide (Lasix<sup>®</sup>)
- K<sup>+</sup>-sparing diuretics: spironolactone, amiloride, triamterene

- First line drug for treatment of mild or moderate (stage 1, 2) essential HT.

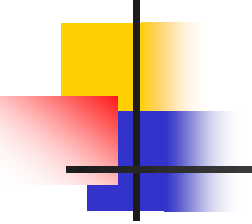
- Increased the efficacy of all other anti-HT



# Diuretics: Side effects

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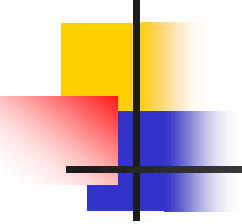
- Electrolyte imbalance- hypokalemia ( $K < 3.5$ ), hypomagnesemia
  - May cause muscle fatigue, cramps, cardiac arrhythmia
- Acid-base imbalance: alkalosis
- Hyperglycemia
- Hyperlipidemia : ↑ LDL, TC, TG
- Orthostatic hypotension
- Hyperuricemia
- Impotence or erectile dysfunction: hydrochlorothiazide
- **All of these SE can be reduced by using low dose diuretics (HCTZ 12.5-25 mg/day)**



# Diuretics: Drug interaction

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- Decrease oral anticoagulant effects
- Digitalis toxicity: arrhythmia
- Amphotericin B and corticosteroid..hypo K
- NSAID, nonselective or selective COX-2 inhibitors: decrease thiazide diuretic effect

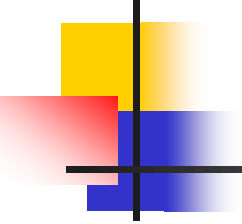


## 2. Sympatholytic agents

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### 2.1 Centrally Acting Sympathoplegic Drugs

- Methyldopa
- Methyldopa  $\longrightarrow$   $\alpha$ -methylnorepinephrine
- Stimulated presynaptic  $\alpha_2$ -adrenergic receptor  $\longrightarrow$   
 $\downarrow$  sympathetic outflow in the brainstem  $\longrightarrow \downarrow$  PVR, HR, CO
- Therapeutic use
  - Methyldopa: hypertension during pregnancy



## 2. Sympatholytic agents

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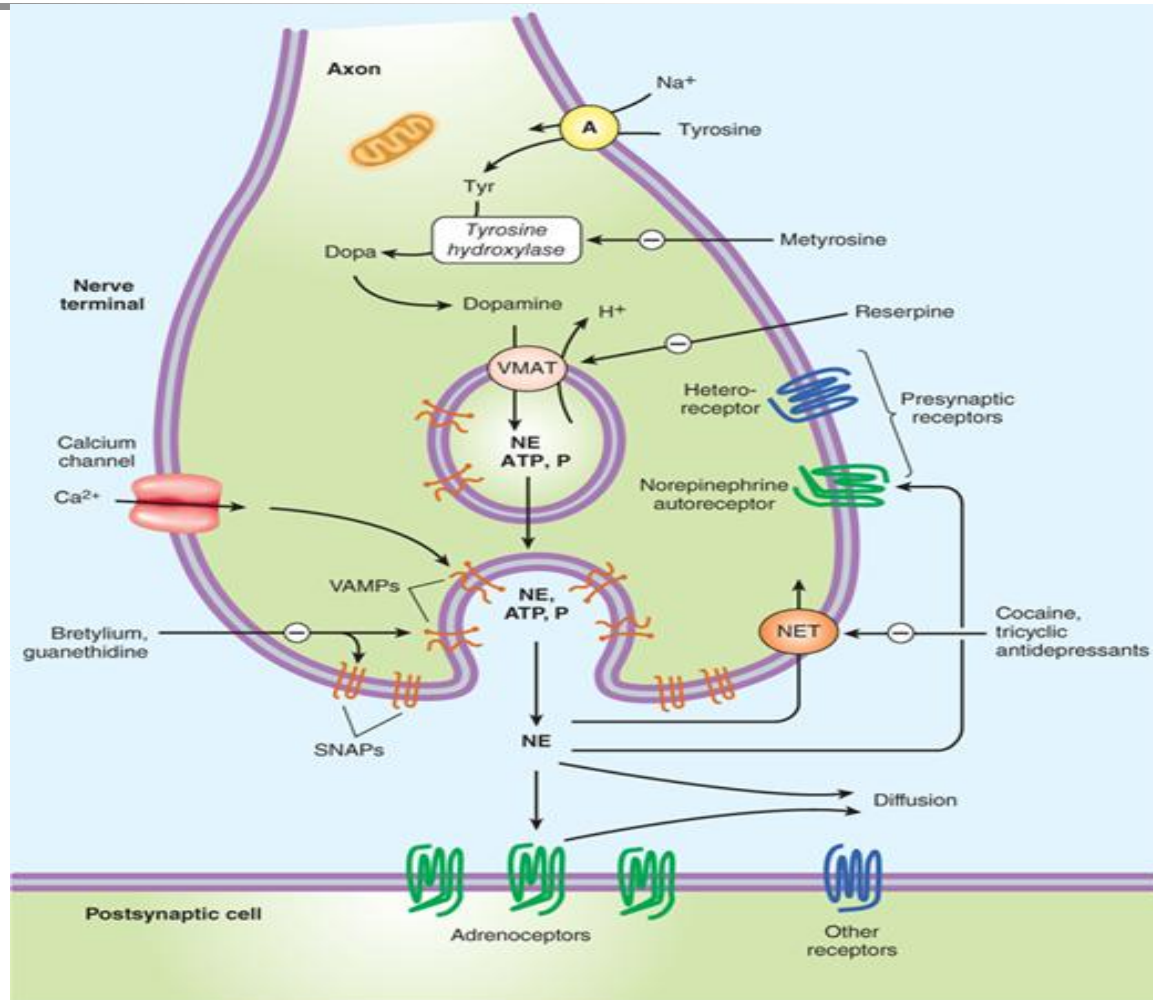
### 2.1 Centrally Acting Sympathoplegic Drugs

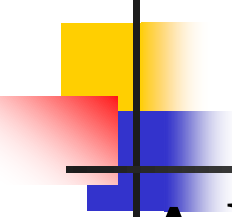
- **Adverse drug reaction:** methyldopa
  - Sedation
  - Dry mouth
  - Vertigo, extrapyramidal signs (parkinsonism signs)
  - Hyperprolactinemia (galactorrhea, gynecomastia)
  - Hemolytic anemia, leukopenia, thrombocytopenia (rare)
  - Hepatitis

## 2. Sympatholytic agents

### 2.2 Postganglionic Neuron-Blocking Agents

- Guanadrel , reserpine
- Combination with diuretics in elderly
- ADR: sedation (common), depression





## **2. Sympatholytic agents**

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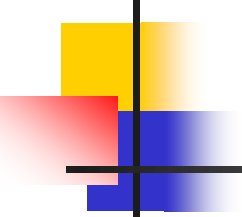
### **2.3 Adrenoceptor antagonists**

#### **2.3.1 Alpha adrenergic receptor blockers ( $\alpha$ -blockers)**

- Selective  $\alpha_1$ -blockers : alfuzosin, doxazosin, prazosin, silodosin, tamsulosin, and terazosin
- Nonselective  $\alpha$ -blockers : phentolamine, phenoxybenzamine

#### **Mechanism of action**

- Selective blocked  $\alpha_1$  receptors in arterioles & venules :  $\downarrow$  PVR

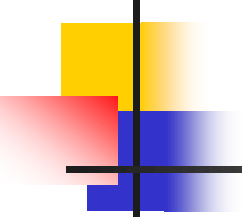


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## **2.3.1 Alpha adrenergic receptor blockers ( $\alpha$ -blockers)**

### **Therapeutic use**

- Not recommend as monotherapy for hypertension
- Use in combination with first-line drug (diuretics,  $\beta$ -blockers ) in elderly male with benign prostatic hypertrophy (BPH)
  - Alpha1-blocker increase urine flow and decrease urinary frequency by inhibiting NE-induced contraction of prostate smooth muscle



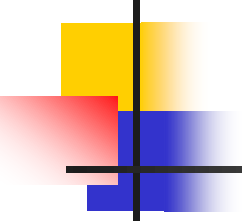
## 2. Sympatholytic agents

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**2.4.1 Alpha-1 adrenergic blockers:** alfuzosin, doxazosin, prazosin, silodosin, tamsulosin, and terazosin

### ■ Side effects

- First-dose phenomenon: postural hypotension, syncope
- Headache, dizziness
- Less reflex tachycardia, palpitation
- Weakness



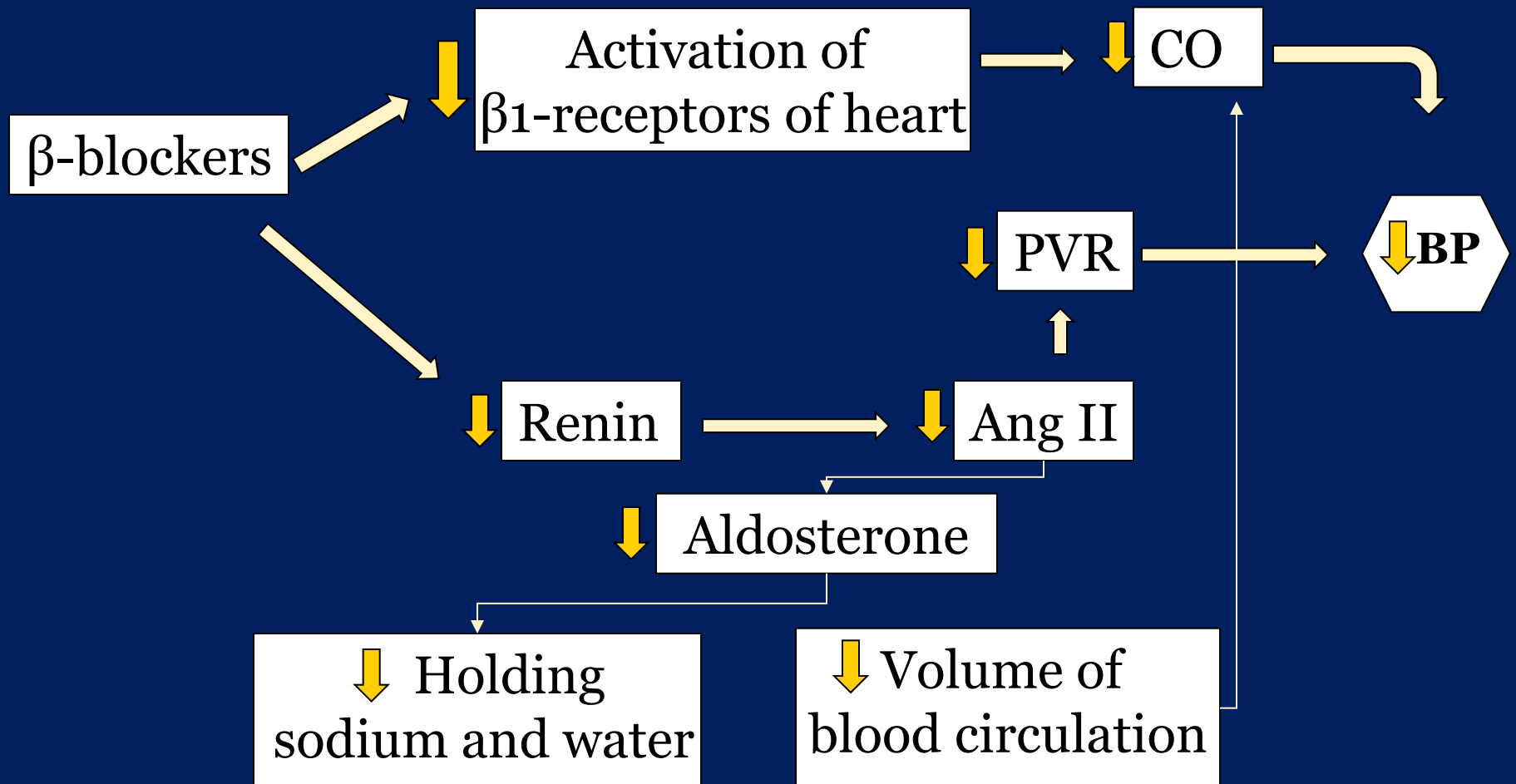
## 2. Sympatholytic agents

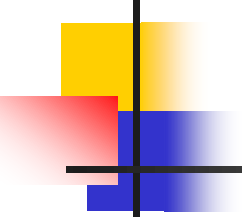
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### 2.4.2 Beta Adrenergic Blocking Agents

- Selective  $\beta_1$ -blockers: metoprolol, atenolol
- Non-selective  $\beta_1$ -blockers: propranolol
- Labetalol :  $\beta$ -blockers with  $\alpha$ -blockers
- Nebivolol :  $\beta$ -blockers with vasodilation (NO pathway)
- Acebutolol :  $\beta$ -blockers with intrinsic sympathomimetic activity (ISA).

# Mechanism of action of $\beta$ -blockers





# Sympatholytic agents

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## 2.4.2 Beta Adrenergic Blocking Agents

### Therapeutic use

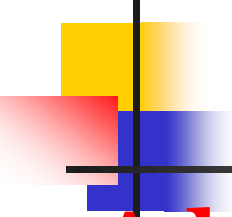
- Sole or combined with diuretic or vasodilators for treatment of hypertension stage 1 or 2

### Adverse drug reaction

- Bradycardia
- Bronchospasm, wheezing, cold extremities
- Hyperglycemia
  - Diabetic: hypoglycemia
- Heart failure: edema, dyspnea

### Contraindication: : Non-selective BB and High dose selective BB

- Asthma, DM, SA and AV nodal dysfunction



# Monitoring for beta-blockers

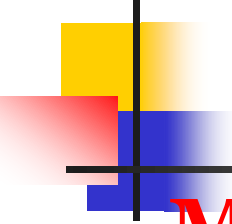
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## Adverse effect

- Bradycardia (resting HR should not <60 bpm)
- Metabolic side effect
  - Blood sugar
  - Lipid profile: ↓HDL, ↑ TG

## Beta-blocker withdrawal

- When BB are suddenly discontinued in pts with ischemic heart disease, angina may be exacerbated
- Taper dose when discontinuing (by 50% for 3 days, and then by another 50% for 3 days)



# Beta blockers

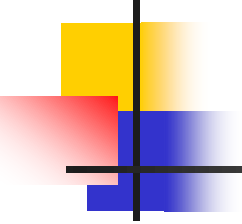
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## MOA

- **Negative chronotropic and negative inotropic activity**

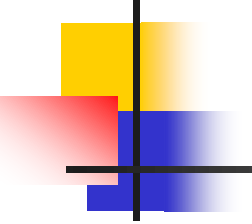
**Favoring used only for HT with**

- **Coronary heart disease**
- **Post-MI**
- **Heart failure**
- **Atrial fibrillation**
- **Pregnancy**



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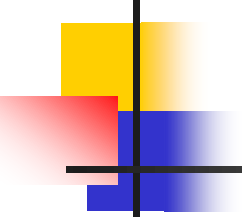
## **3. Vasodilators**



# Direct Acting Vasodilators

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- **Oral vasodilator:** hydralazine, minoxidil
- **Parenteral vasodilators:** nitroprusside, diazoxide, fenoldopam (for hypertensive emergencies)
- **Therapeutic use:**
  - Hydralazine: combination with nitrates: both HT and heart failure
  - Minoxidil: combination with  $\beta$ -blocker & loop diuretic: severe HT
  - Nitroprusside: HT emergencies



# Mechanism of action and ADR

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- Hydralazine
  - Arterial vasodilator
  - Lupus erythematosus-like syndrome
  - Headache, nausea, anorexia, palpitation, sweating and flushing
- Minoxidil
  - Arterial vasodilators by opening of potassium channels... very potent
  - Tachycardia, palpitation, hypertrichosis
- Sodium nitroprusside
  - Dilate both artery and vein by activation of guanylyl cyclase by release of NO
  - Cyanide accumulation, metabolic acidosis
- Diazoxide
  - Arterial vasodilator by opening of potassium channel
  - Excessive hypotension, angina pectoris
- Fenoldopam
  - D1 agonist-arterial dilators
  - Tachycardia, headache and flushing



- **Directly acting vasodilators**

- **Drugs that activate potassium channels.. minoxidil, cromokalim, lemakalim , nicorandil ( plus NO donor activity)**
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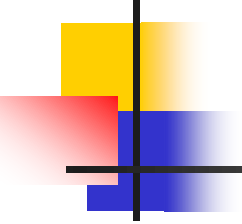
- **Drugs that act via cyclic nucleotides**

- **Natriuretic peptides: vasodilator/ activation of guanylate cyclase: Anaritide**

- **Candoxatril: a neutral endopeptidase inhibitor**
- **Omapatrilat: a combined neutral endopeptidase inhibitor and ACEI**

- **Indirectly acting vasodilator drugs**

- **Imidazoline I<sub>1</sub> receptor agonists at RVLM: moxonidine, rilmenidine**
- **Endothelin receptor antagonist: Bosentan, aprocitentan**



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## **4. Calcium Channel Blocking Drugs**

# Calcium Channel Blockers (CCBs)

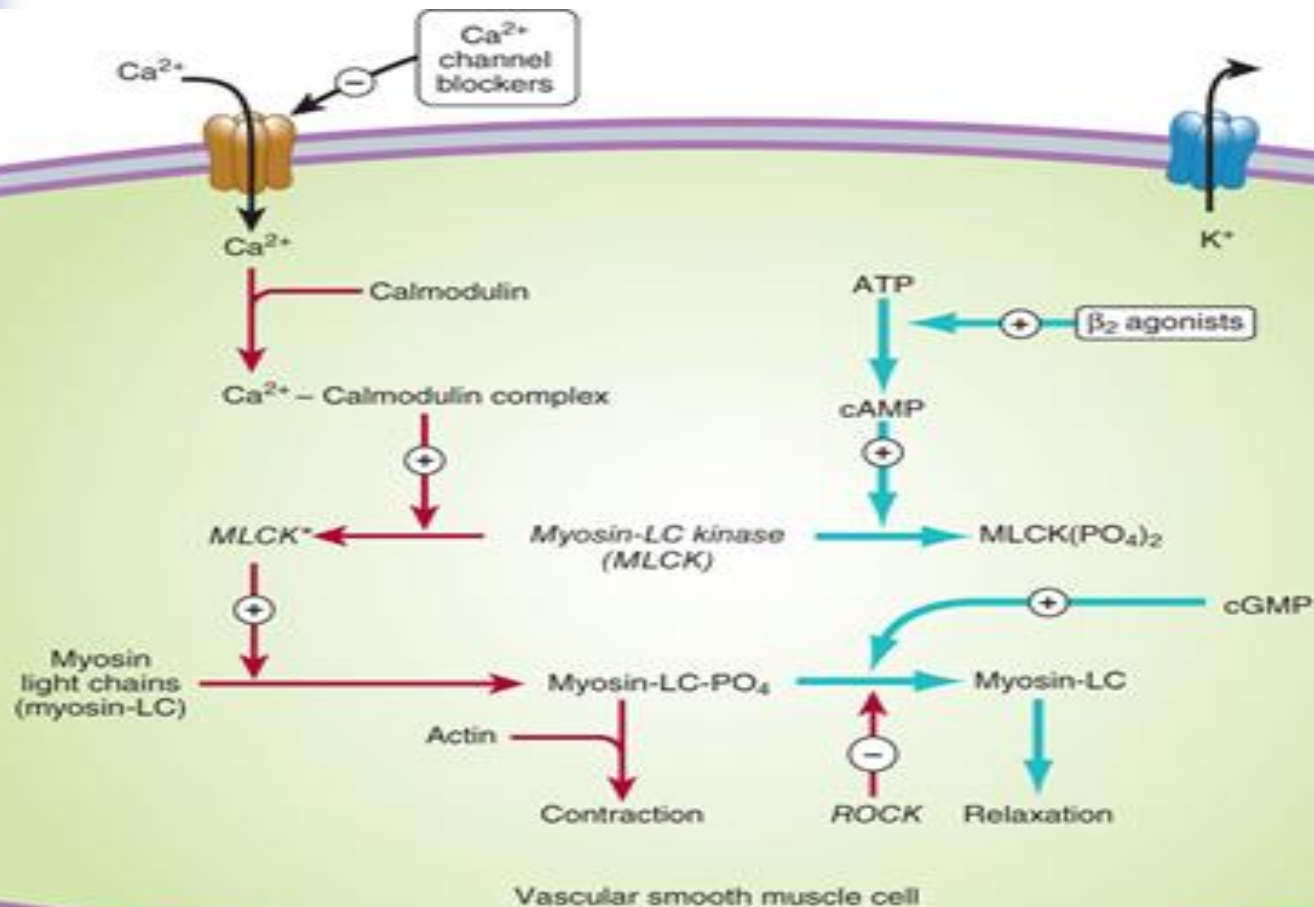
## DIHYDROPYRIDINES:

- ❖ *Amlodipine, Nifedipine, Nicardipine etc*
- ❖ *bind with  $\alpha_1$ - subunit (N- sites) of L-type  $\text{Ca}^{2+}$  channel*
- ❖ *vascular smooth muscle > myocardium*
- ❖ *peripheral or coronary vasodilatation*

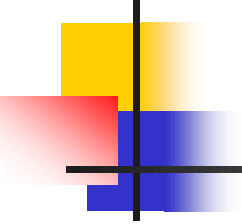
## NONDIHYDROPYRIDINES:

- ❖ *Verapamil, Diltiazem*
- ❖ *bind with  $\alpha_1$ - subunit (N- sites) of L-type  $\text{Ca}^{2+}$  channel*
- ❖ *myocardium > vascular smooth muscle*
- ❖ *decrease HR and contractility*

# CCBs – Mechanisms of Action



# Calcium Channel Blockers (CCBs)

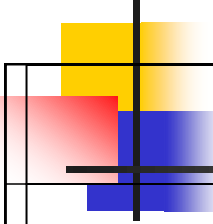


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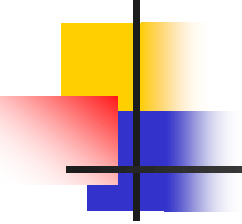
## Therapeutic use

- Used alone or combination with other drugs (diuretic),  $\beta$ -blockers are contraindication
- Preferred treatment with isolated systolic hypertension

# Comparative Adverse Effects



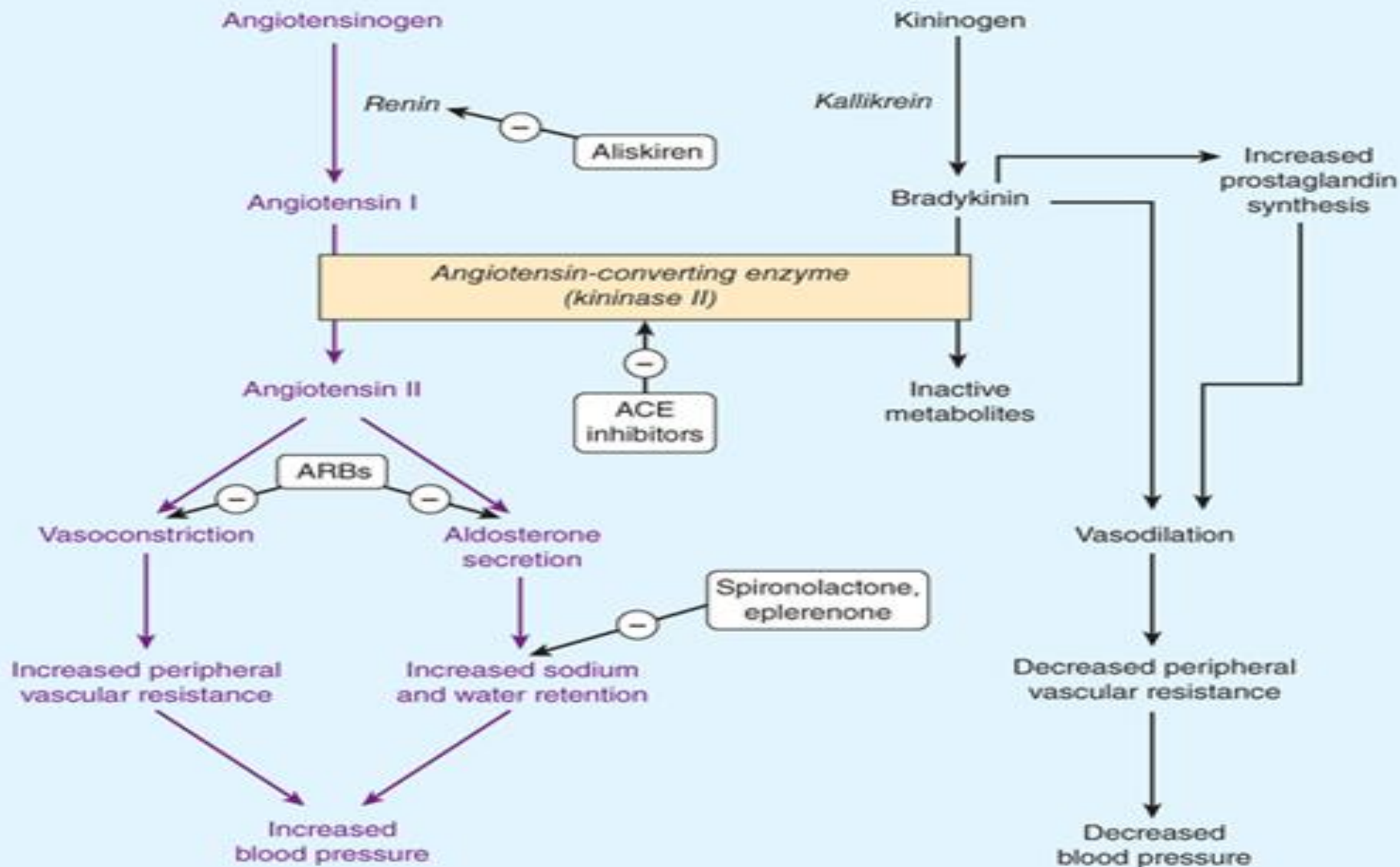
|                         | Diltiazem | Verapamil | Dihydropyridines |
|-------------------------|-----------|-----------|------------------|
| Overall                 | 0-3%      | 10-14%    | 9-39%            |
| Hypotension             | ++        | ++        | +++              |
| Headaches               | 0         | +         | +++              |
| Peripheral Edema        | ++        | ++        | +++              |
| Constipation            | 0         | ++        | 0                |
| CHF (Worsen)            | 0         | +         | 0                |
| AV block                | +         | ++        | 0                |
| Caution w/beta blockers | +         | ++        | 0                |



## 5. Renin angiotensin aldosterone inhibitors

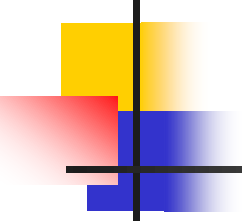
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1. Angiotensin-converting enzyme (ACE) inhibitors
2. Angiotensin receptor blockers (ARBs)
3. Direct renin inhibitors
4. Aldosterone synthase inhibitors
5. Angiotensin receptor neprilysin inhibitors (ARNIs)



Source: Bertram G. Katzung, Anthony J. Trevor: Basic & Clinical Pharmacology, 13th Ed.  
www.accesspharmacy.com  
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## SITE OF ACTION OF DRUGS

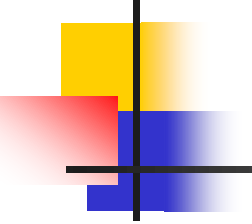


## 5.1 Angiotensin-converting enzyme (ACE) inhibitors

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### Classification of ACE inhibitors

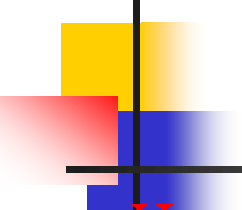
- **Sulfhydryl containing ACE inhibitors:** captopril, fentiapril, pivalopril, zofenopril, alacepril
- **Dicarboxyl containing ACE inhibitors:** lisinopril, benazepril, quinapril, cilazapril
- **Phosphorus containing ACE inhibitors:** fosinopril



## 5.1 ACE inhibitors

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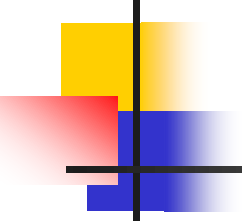
- Suppress Ang II formation: vasodilation
- Decrease aldosterone release: ↓ Na & water reabsorption
- Increase bradykinin level: decreased PVR
- Difference in potency, pharmacokinetics, active form
- Similar in indications and adverse effects
- 1<sup>st</sup> line therapy in HT stage 1, 2 with DM, LVH, post MI



# ACE inhibitors: Adverse effects

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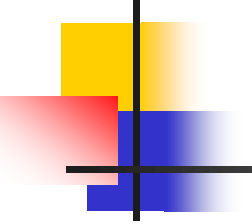
- Hypotension
- Cough
  - Mediated by accumulation of bradykinin, substance P and /or prostaglandins
- Hyperkalemia
- Acute renal failure
- Fetopathic potential: second and third trimester (death ,pulmonary hypoplasia)
- Angioneurotic edema: not dose related ? Bradykinin release
- Skin rash
- Proteinuria
- Dysgeusia
- Neutropenia
- Glucosuria
- Hepatotoxicity



# ACEIs: Contraindication

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- Pregnancy
  - Fatal hypotension, anuria, renal failure and developmental deformities
- Previous angioneurotic oedema
- Bilateral renal artery stenosis
- Renal artery stenosis in solitary kidney

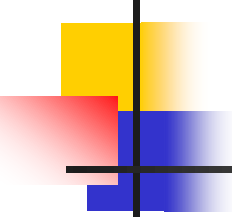


# Drug interactions with ACE inhibitors

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- Antacids...reduce bioavailability
- Capsaicin....cough
- NSAIDS... reduce effects
- K<sup>+</sup>-sparing diuretics...hyperkalemia

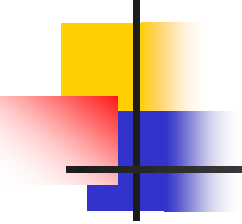
## 5.2 Angiotensin II receptor blockers (ARBs)



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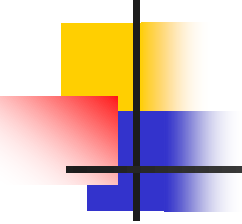
- Losartan, Valsartan, Candesartan, Eprosartan, Irbesartan, Telmisartan
- Inhibit AT<sub>1</sub> receptor: vasodilation
- Decrease aldosterone release: ↓ Na & water reabsorption
- Less effects on angiotensin and bradykinin level
- Therapeutic use
  - Monotherapy in hypertension stage 1,2
  - Angiotensin dependent hypertension

# ADR of angiotensin II receptor blockers (ARBs)



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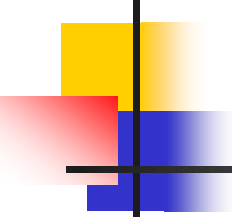
- No cough
- Less angioneurotic edema
- Fetopathic potential
- Hypotension
- Acute renal failure
- Hyperkalemia



# ARBs: Contraindication

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- Pregnancy
  - Fatal hypotension, anuria, renal failure and developmental deformities
- Previous angioneurotic oedema
- Bilateral renal artery stenosis
- Renal artery stenosis in solitary kidney

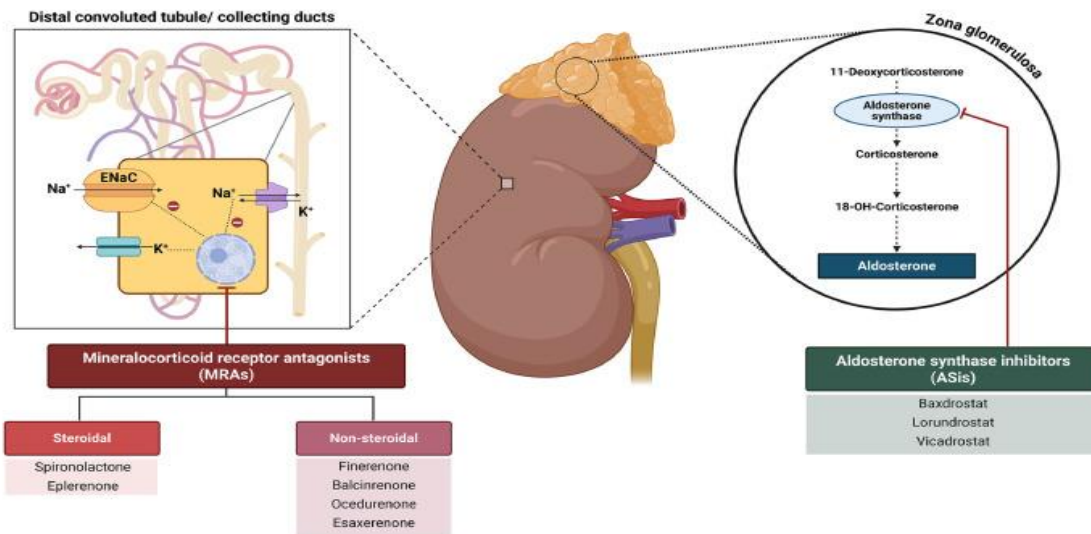


## 5.3 Direct renin inhibitor

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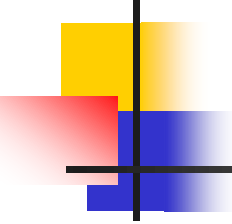
- Potent competitive inhibitor of renin
- Reduced angiotensin I synthesis  $\longrightarrow$   $\downarrow$  Ang II
- Low bioavailability, plasma protein binding 50%,  $t_{1/2}$  20–45 h, metabolism by CYP3A4
- Aliskiren has blood pressure–lowering effects similar to those of ACE inhibitors and ARBs.
- SE: hyperkalemia, diarrhea, increase BUN and creatinine level

# 5.4 Aldosterone Synthase Inhibitors and Mineralocorticoid Receptor Antagonists



| MRA vs ASi effect in key prognostic markers |                                                                                                                                |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Blood pressure                              | Similar effects according to available evidence although limited by the absence of adequately powered head-to-head comparisons |
| Serum potassium                             |                                                                                                                                |
| eGFR                                        |                                                                                                                                |
| UACR                                        |                                                                                                                                |

- Decrease in aldosterone effects
- **Therapeutic use:** Hypertension inadequately controlled on other drugs
- **Side effects:** elevated blood potassium, Gastrointestinal issues, hormonal and sexual effects (Spironolactone), Reduced sexual side effects (Eplerenone and Finerenone)



## 5.5 Angiotensin receptor neprilysin inhibitors (ARNIs)

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- Sacubitril-Valsartan
- Sacubitril is a pro-drug that, upon activation, acts as a **neprilysin inhibitor**.
- Valsartan is an ARB that works by blocking the RAAS system.
- FDA-approved to treat patients with chronic heart failure with reduced ejection fraction (HFrEF).
- **Adverse effects** include hypotension, hyperkalemia, renal failure, cough, and angioedema.

# Long term effects of antihypertensive drugs

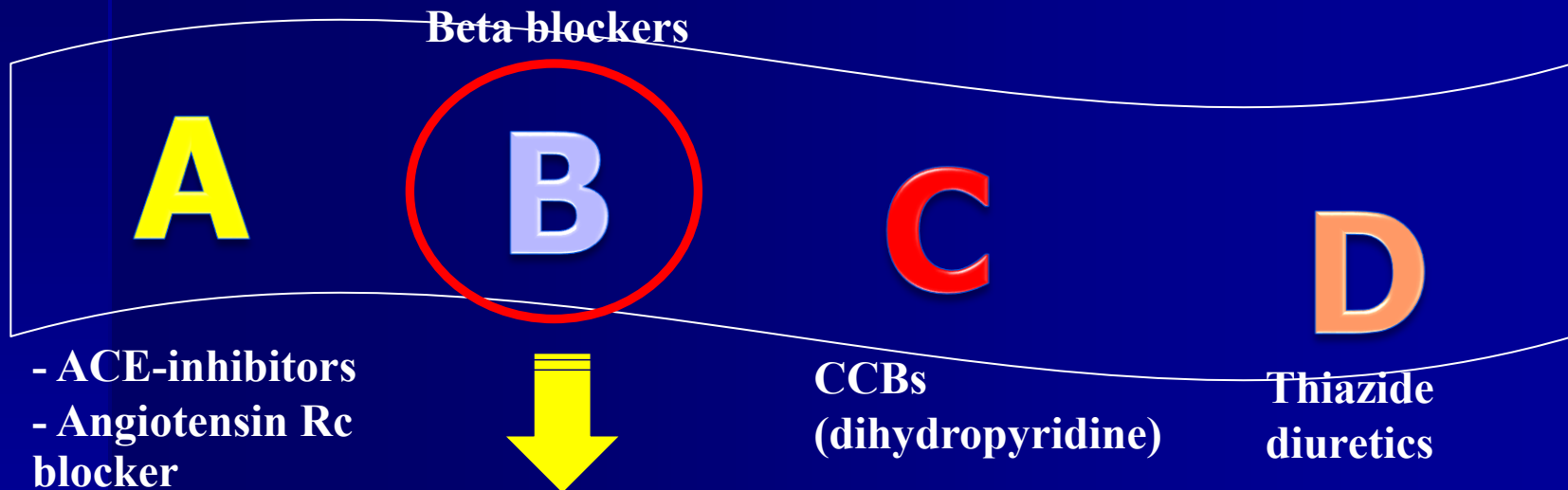
|                                | Heart rate | Cardiac output | Total peripheral resistance | Plasma volume | Plasma renin activity |
|--------------------------------|------------|----------------|-----------------------------|---------------|-----------------------|
| <b>Diuretics</b>               | ↔          | ↔              | ↓                           | −↓            | ↑                     |
| <b>Sympatholytic agents</b>    |            |                |                             |               |                       |
| <b>Centrally acting</b>        | −↓         | −↓             | ↓                           | −↑            | −↓                    |
| <b>α1 Blockers</b>             | −↑         | −↑             | ↓                           | −↑            | ↔                     |
| <b>β Blockers</b>              |            |                |                             |               |                       |
| <b>No ISA</b>                  | ↓          | ↓              | −↓                          | −↑            | ↓                     |
| <b>ISAa</b>                    | ↓↑         | ↔              | ↓                           | −↑            | −↓                    |
| <b>Arteriolar vasodilators</b> | ↑          | ↑              | ↓                           | ↑             | ↑                     |
| <b>CCBs</b>                    | ↓ or ↑     | ↓ or ↑         | ↓                           | −↑            | −↑                    |
| <b>ACEIs</b>                   | ↔          | ↔              | ↓                           | ↔             | ↑                     |
| <b>ARBs</b>                    | ↔          | ↔              | ↓                           | ↔             | ↑                     |
| <b>Renin inhibitor</b>         | ↔          | ↔              | ↓                           | ↔             | ↓ (but renin ↑)       |

# Drug treatment strategy for HTN

2025 ACC/AHA Guideline, 2024 ESC/ESH Guideline

- Surrogate endpoint: Blood pressure
- Clinical endpoint: Morbidity and mortality

## First-line drug treatment



*"Consider when there is a specific indication e.g. ischemic heart disease, heart failure, atrial fibrillation or younger women with, or planning, pregnancy"*

# Compelling and possible contra-indications to the use of antihypertensive drugs

## 2024 ESC/ESH Guideline

| Drug                                       | Compelling                                                                    | Possible                                                                               |
|--------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Diuretic (thiazide)                        | Gout                                                                          | Metabolic syndrome<br>Glucose intolerance<br>Pregnancy<br>Hypercalcemia<br>Hypokalemia |
| Beta-blockers                              | Asthma<br>High grade SA or AV block<br>Bradycardia (HR<60 bpm)                | Metabolic syndrome<br>Glucose intolerance<br>Athletes and physically active pts        |
| Calcium antagonists (dihydropyridines)     |                                                                               | Tachyarrhythmia<br>Heart failure<br>Pre-existing severe leg oedema                     |
| Calcium antagonists (verapamil, diltiazem) | High grade SA or AV block<br>Severe LV dysfunction<br>Bradycardia (HR<60 bpm) | Constipation                                                                           |

# Compelling and possible contra-indications to the use of antihypertensive drugs

## 2024 ESC/ESH Guideline

| Drug                                   | Compelling                                                                                    | Possible                          |
|----------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------|
| ACE inhibitors                         | Pregnancy<br>Previous angioneurotic oedema<br>Hyperkalemia<br>Bilateral renal artery stenosis | Women with childbearing potential |
| Angiotensin receptor blockers          | Pregnancy<br>Hyperkalemia<br>Bilateral renal artery stenosis                                  | Women with childbearing potential |
| Mineralocorticoid receptor antagonists | Acute or severe renal failure (eGFR<30 ml/min)<br>Hyperkalemia                                |                                   |

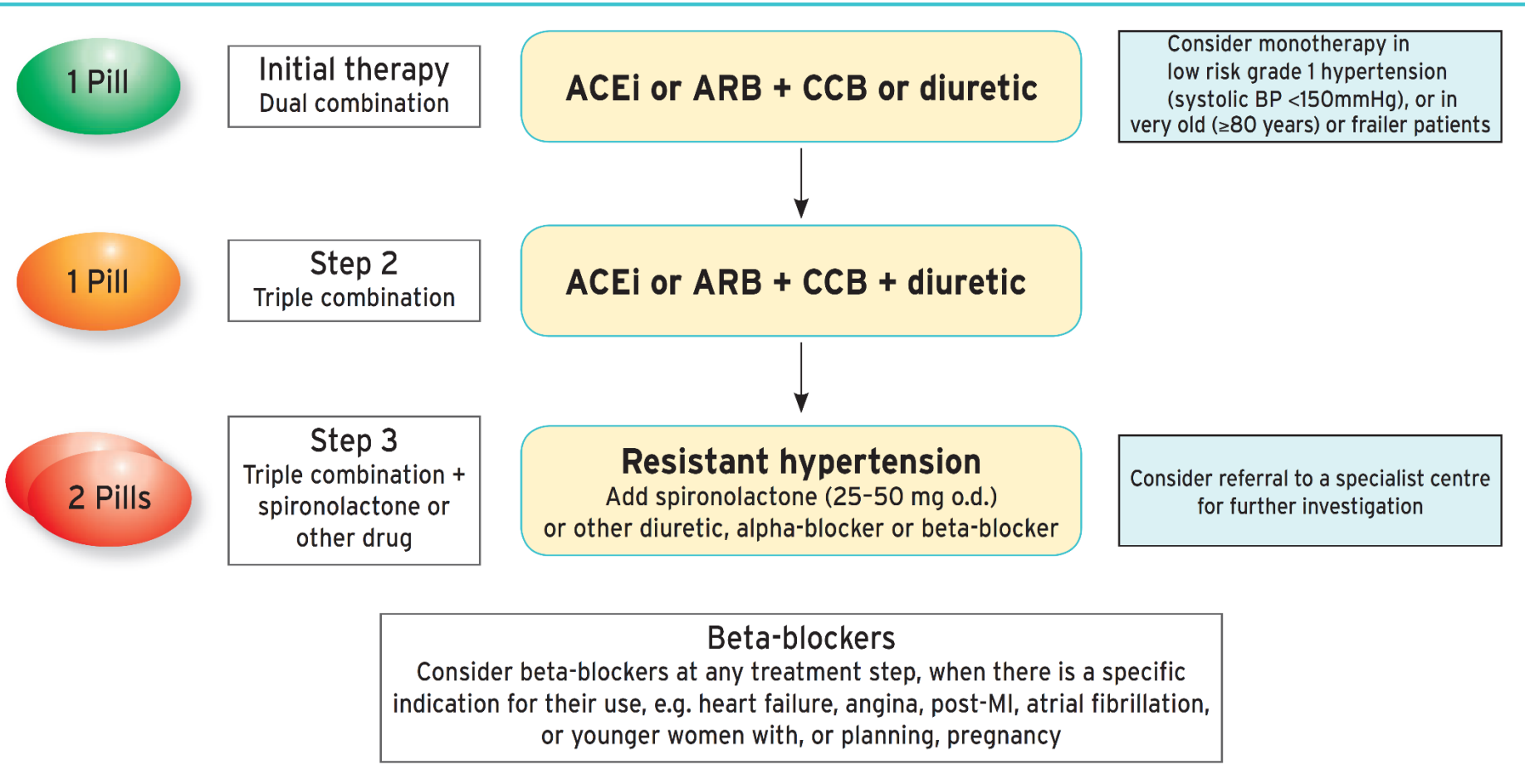
# Drug treatment strategy for HTN

## 2024 ESC/ESH Guideline

- **Two-drug combination** is recommended as initial therapy **except** in frail older patients and those at low risk with grade 1 HTN (esp. if SBP <150 mmHg)
  - Preferred combination: **RAS blockers (either ACEIs or ARB) with CCB or Diuretic**
  - Preferably in a **single-pill combination (SPC)**
  - **The combination of two RAS blockers is not recommended**

# Core drug treatment strategy for uncomplicated HTN

## 2024 ESC/ESH Guideline



# **Drug treatment strategy for HTN with other comorbidities**

**2024 ESC/ESH Guideline**

**CORONARY ARTERY DISEASE**

**CHRONIC KIDNEY DISEASES**

**SECONDARY STROKE PREVENTION**

**HEART FAILURE**

**ATRIAL FIBRILLATION**

# Questions ?

